

- Although a possible source of SI breakdown, residual foreground contamination would need to be of order the intrinsic CMB temperature anisotropy to account for an appreciable effect [32–34].
- It would be erroneous to assume that the true CMB temperature fluctuations are completely extracted from the observed map. Observational artifacts such as non-circular beam, inhomogeneous noise correlation, residual striping patterns could be potential sources of SI breakdown.

Violation of statistical isotropy of CMB anisotropy and its measurement has been discussed in literature earlier [35–40] by defining an estimator where SI breakdown in an observed CMB anisotropy sky map is indicated by non zero value of this estimator. Studies have also been done by implementing a directional dependent inflationary power spectrum  $P(\vec{k})$  which gives rise to off-diagonal terms in the covariance matrix [41], [42]; spontaneous breakdown of SI in the CMB by a non-linear response to long-wavelength field fluctuations that appear as a gradient locally to the observer [43] or locally through a modulation field [44]; incorporating an initial period of kinetic energy domination in single field inflation [45].

In this paper we present a new formulation that relates the breakdown of SI in the CMB photon fluctuations at last scattering, and evolving them to find the effect of the modes at present epoch hence the CMB  $\langle a_{\ell m} a_{\ell' m'} \rangle$  today. We also find the Bipolar spherical harmonic coefficients (BipoSH) [38, 39] which are linear combinations of off-diagonal elements of the covariance matrix. BipoSH expansion completely represents the information of the covariance matrix thus being the most general way of studying two point correlation functions of CMB anisotropy. These BipoSH coefficients are mathematically complete measures of SI violation on a sphere.

## II. REVIEW OF STATISTICALLY ISOTROPIC CMB BRIGHTNESS FLUCTUATIONS

### A. Boltzmann equations, inhomogeneities and anisotropies

In the smooth background universe, thermalized photons being distributed homogeneously and isotropically, the temperature  $T$  is independent of  $\vec{x}$  and direction of propagation  $\hat{p}$  respectively. To describe perturbations about this smooth universe, we allow inhomogeneities in the photon distribution and anisotropies.

Before recombination,  $z_{\text{rec}} \approx 1100$ , the photons were tightly coupled to the electrons and protons; all together they can be described as a single fluid, the baryon-photon fluid. After recombination, photons free-stream from the surface of last scattering to the present epoch.

Given the cosmological perturbations to the photons at recombination, one can predict the anisotropy spectrum today. The main motivation next is to relate the

moments today to the moments at recombination using the photon distribution function.

### B. Fluctuations of CMB photon distribution

In the Boltzmann equation for photons  $df/dt = C[f]$ , we expand the photon distribution function  $f(\vec{x}, p, \hat{n}, \tau)$  about its zero-order Bose-Einstein value  $T(\tau)$  [46] where  $\vec{p} = p\hat{n}$ . The distribution function of the photons changes with the perturbed temperature as

$$f(\vec{x}, p, \hat{n}, \tau) = \left[ \exp \left\{ \frac{p}{T(\tau)[1 + \Delta(\vec{x}, \hat{n}, \tau)]} \right\} - 1 \right]^{-1}. \quad (1)$$

The perturbation to the distribution function is characterized by  $\Delta \equiv \delta T/T$  termed as CMB brightness fluctuations henceforth. Since the perturbation  $\Delta$  is small, we can expand  $f(\vec{x}, p, \hat{n}, \tau)$  keeping only terms up to first order to get

$$\Delta(\vec{x}, \hat{n}, \tau) \equiv \left( \frac{\partial f^0}{\partial \ln p} \right)^{-1} \delta f, \quad (2)$$

where  $f^0$  is the zero-order photon distribution function.  $\Delta(\vec{x}, \hat{n}, \tau)$  depends on  $\vec{x}, \hat{n}$  and  $\tau$  and not on the magnitude of momentum  $p$ ; this is a valid assumption since the temperature of the plasma is very small compared to the rest energy of the electrons which undergo scattering, elastic Thomson scattering has negligible effect on the magnitude of the photon momentum.

Perturbations to the CMB remain small at all cosmological epochs; evolution of the largest scales being in the linear regime. In solving the linear evolution equations, it is simplest to work with Fourier transforms since every Fourier mode evolves independently.

$$\Delta(\vec{x}, \hat{n}, \tau) = \int \frac{d^3 k}{(2\pi)^3} e^{i\vec{k} \cdot \vec{x}} \tilde{\Delta}(\vec{k}, \hat{n}, \tau) \phi(\vec{k}), \quad (3)$$

where  $\phi(\vec{k})$  is the primordial density fluctuations.

With statistical isotropy assumption

$$\begin{aligned} \tilde{\Delta}(\vec{k}, \hat{n}, \tau) &= \tilde{\Delta}(k, \hat{k} \cdot \hat{n}, \tau) \\ &= \sum_{\ell} (-i)^{\ell} (2\ell + 1) \tilde{\Delta}_{\ell}(k, \tau) P_{\ell}(\hat{k} \cdot \hat{n}) \\ &= 4\pi \sum_{\ell m} (-i)^{\ell} \tilde{\Delta}_{\ell}(k, \tau) Y_{\ell m}(\hat{k}) Y_{\ell m}(\hat{n}). \end{aligned} \quad (4)$$

$\tilde{\Delta}_{\ell}(k, \tau)$  are the moments of the CMB brightness fluctuation. The monopole  $\tilde{\Delta}_0$  is related to the density perturbations while the dipole  $\tilde{\Delta}_1 \propto \hat{n} \cdot \vec{v}$ , gives the velocity term for baryons.